## **Lecture 25** (*April 21*)

*Reading: Section 9.1*
*Viewing: none*

Consider the problem of diagonalizing the matrix:

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{208}$$

As we have seen in Fact 19, the recipe for doing so is to find the eigenvalues and eigenvectors of $A$ first. Now, the eigenvalues are the roots of the characteristic polynomial:

$$p(\lambda) = \det \begin{bmatrix} -\lambda & -1 \\ 1 & -\lambda \end{bmatrix} = \lambda^2 + 1 \tag{209}$$

You may be tempted to believe that this polynomial has no roots, because after all, no real number has the property that its square is $-1$. And your guess would seem to be justified: the linear transformation corresponding to the matrix (208) is rotation by 90 degrees of the plane. If the matrix had an eigenvector, then rotation by 90 degrees would have to scale along the direction of that eigenvector, which does not seem to make any sense geometrically.

However, in math, a paradox is rarely the end of the story, but more often it is the catalyst for new discoveries. In the example at hand, the way to fix the fact that the polynomial (209) has no solutions is to add these solutions by hand. What this means is that we are no longer satisfied with real numbers, and instead we want to consider the **imaginary number**:

$$i \quad \text{defined such that} \quad \boxed{i^2 = -1} \tag{210}$$

which is often abbreviated as saying $i = \sqrt{-1}$. You should read this as saying that $i$ is <u>defined</u> to be the square root of $-1$, and it certainly is not a real number. But it is a very useful tool for defining **complex numbers**, which are by definition all expressions of the form:

$$z = a + bi \tag{211}$$

for any real numbers $a$ and $b$. You may ask: why don't we also throw in various powers of $i$ into the expression (211) while we're at it? The reason is that doing so would be redundant, since the powers of $i$ can themselves be expressed in the form (211):

$$i^2 = -1 \qquad i^3 = -i \qquad i^4 = 1 \qquad i^5 = i \qquad i^6 = -1 \qquad \ldots \tag{212}$$

**Remark.** *I hope you won't be put off by the feeling that complex numbers are "contrived". In the past, people have argued for hundreds of years whether negative integers make any sense, and ultimately we accepted them. And if you think about it, what is the meaning of "$-2$" other than "the thing to which you add 2 to get 0" and what is the meaning of "$\sqrt{3}$" other than "the thing which you square to get 3". Similarly, "$i$" is defined as "the thing which you square to get $-1$".*

Complex numbers can be added and multiplied using the usual algebra operations:

$$(a + ib) + (c + id) = (a + c) + i(b + d) \tag{213}$$

$$(a+ib)(c+id) = ac + adi + bci + bdi^2 = (ac-bd) + (ad+bc)i \tag{214}$$

so the sum and product of complex numbers is also a complex number. But complex numbers have other kinds of operations associated to them. If $z = a + bi$ is a complex number, then:

$$\text{Re } z = a \text{ is called the } \textbf{real part} \text{ of } z \tag{215}$$
$$\text{Im } z = b \text{ is called the } \textbf{imaginary part} \text{ of } z \tag{216}$$
$$\bar{z} = a - bi \text{ is called the } \textbf{conjugate} \text{ of } z \tag{217}$$

Therefore, the conjugate of a complex number is that complex number with the same real part, but the opposite imaginary part. Moreover, define the **absolute value** of a complex number as:

$$|z| = \sqrt{a^2 + b^2} \tag{218}$$

If this looks familiar, then it should! Since a complex number (211) is just a pair of real numbers, it encodes the same information as a point in the plane, or a vector in $\mathbb{R}^2$. Then the absolute value of the complex number is just the length of the corresponding vector. Let us prove that:

$$\boxed{z\bar{z} = |z|^2} \tag{219}$$

Indeed, this is because:

$$(a+bi)(a-bi) = a^2 - abi + abi - b^2i^2 = a^2 + b^2$$

Formula (219) is important because it allows us to write:

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2} \qquad \text{i.e.} \qquad \frac{1}{a+bi} = \frac{a-bi}{a^2+b^2} \tag{220}$$

and this is great because it allows us to divide complex numbers:

$$\frac{2+i}{4+3i} = \frac{(2+i)(4-3i)}{4^2+3^2} = \frac{8-6i+4i-3i^2}{25} = \frac{11}{25} - \frac{2}{25}i$$

**Example 7.** *We now have the tools to diagonalize the matrix* (208)*. Its characteristic polynomial has two roots, which are complex numbers:*

$$p(\lambda) = \lambda^2 + 1 = (\lambda - i)(\lambda + i) \qquad \textit{has roots} \qquad \begin{cases} \lambda_1 = i \\ \lambda_2 = -i \end{cases}$$

*so the eigenvalues are $i$ and $-i$. Now we need to find eigenvectors corresponding to these eigenvalues:*

$$\begin{aligned} A\boldsymbol{v}_1 = i\boldsymbol{v}_1 \quad &\Rightarrow \quad (A - iI)\boldsymbol{v}_1 = 0 \quad \Rightarrow \quad \boldsymbol{v}_1 \in N(A - iI) \\ A\boldsymbol{v}_2 = -i\boldsymbol{v}_2 \quad &\Rightarrow \quad (A + iI)\boldsymbol{v}_2 = 0 \quad \Rightarrow \quad \boldsymbol{v}_2 \in N(A + iI) \end{aligned}$$

*To compute the nullspaces in question, we perform our good friend Gaussian elimination:*

$$A - iI = \begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix} \quad \overset{REF}{\rightsquigarrow} \quad \begin{bmatrix} -i & -1 \\ 0 & 0 \end{bmatrix}$$

*just by adding* $(-i)$ *times the first row to the second row. Therefore,* $\boldsymbol{v}_1 = \begin{bmatrix} x \\ y \end{bmatrix}$ *lies in* $N(A - iI)$ *if:*

$$\begin{bmatrix} -i & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \quad \Rightarrow \quad -ix - y = 0$$

*so a choice of eigenvector for the eigenvalue* $i$ *is* $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$. *Similarly, one finds that a choice of eigenvector for the eigenvalue* $-i$ *is* $\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ i \end{bmatrix}$. *Therefore, Fact 19 implies that:*

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = V \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} V^{-1} \qquad \textit{where} \qquad V = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix}$$

*and this concludes the diagonalization of the matrix* $A$ *of* (208).

Any real number can also be interpreted as a complex number, with imaginary part 0. With this in mind, we note that any quadratic polynomial has two complex roots, specifically:

$$ax^2 + bx + c = 0 \qquad \text{has solutions} \qquad \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{221}$$

If $b^2 - 4ac \geq 0$, the solutions are real numbers. However, if $b^2 - 4ac < 0$, the square root in formula (221) will be an imaginary number (i.e. a multiple of $i$) and the solutions are complex non-real numbers. What's more, it turns out that complex numbers allow us to find roots of arbitrary polynomials. To this end, the **fundamental theorem of algebra** states that:

$$\textbf{any polynomial of degree } n \textbf{ has exactly } n \textbf{ complex roots} \tag{222}$$

where the word "exactly" should be taken to mean "counted with multiplicities", as in Lecture 23.

Let's discuss one more feature of complex numbers, which is very useful in computations: the polar form. As we have seen, a complex number can be represented as a point in the plane via **Cartesian coordinates**:

$$z = a + ib \qquad \rightsquigarrow \qquad (a, b)$$

However, we can also represent the same point via **polar coordinates**:

$$z = a + ib \qquad \rightsquigarrow \qquad (r, \theta)$$

where $r = \sqrt{a^2 + b^2}$ is the absolute value of $z$ (i.e. the size of a circle around the origin where the point is located) and $\theta \in [0, 2\pi)$ is the angle between the point in question and the horizontal line:

$$\cos\theta = \frac{a}{\sqrt{a^2 + b^2}} \qquad \Rightarrow \qquad \theta = \arccos\left(\frac{a}{\sqrt{a^2 + b^2}}\right)$$

The angle $\theta$ is called the **argument** of $z$, and is only defined up to adding arbitrary integer multiples of $2\pi$ (because $\theta$ and $\theta + 2\pi$ determine the same angle, there is an inherent ambiguity in the notion of "argument"). Since $(\cos\theta)^2 + (\sin\theta)^2 = 1$, we have $\sin\theta = \frac{b}{\sqrt{a^2+b^2}}$ and therefore:

$$\boxed{z = a + bi = r(\cos\theta + i\sin\theta)} \tag{223}$$

The above is called the **polar form** of $z$. Recall the Taylor series expansions of $\cos\theta$ and $\sin\theta$:

$$\cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \dots$$
$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \dots$$

which implies that:

$$\cos\theta + i\sin\theta = 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{i\theta^5}{5!} - \dots$$

The right-hand side is the Taylor series of $e^{i\theta}$ (see (212)), so we obtain the following identity:

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta} \tag{224}$$

The polar form (223) of a complex number can therefore be written in a nicer form as:

$$\boxed{z = a + bi = re^{i\theta}} \tag{225}$$

Since $|e^{i\theta}| = 1$, the polar form of a complex number can be said to separate its absolute value part (namely $r$) from its angular part (namely $e^{i\theta}$). One of the good things about the polar form is that, as opposed from the Cartesian form, it's handy for multiplying numbers and raising to powers:

$$z^n = r^n e^{in\theta} \tag{226}$$

and:

$$zz' = rr'e^{i(\theta+\theta')} \tag{227}$$

if $z = re^{i\theta}$ and $z' = r'e^{i\theta'}$. In other words, we have:

$$|zz'| = |z||z'|$$
$$\arg(zz') = \arg(z) + \arg(z')$$

(the sum of arguments formula only holds up to adding integer multiples of $2\pi$). Formula (226) allows us to describe the roots of unity, i.e. solutions of:

$$z^n = 1 \tag{228}$$

If $z$ is a real number, the only solution to this equation is $z = 1$ (and also $z = -1$ if $n$ is even), but the equation in question has more solutions in the world of complex numbers. Specifically, as a consequence of the fundamental theorem of algebra (222), we expect the equation (228) to have $n$ complex solutions. If $z = re^{i\theta}$, then combining formulas (226) and (228) gives us:

$$r^n e^{in\theta} = 1$$

Since $r$ is a positive real number, we must have $r = 1$. Since $\theta$ is an angle, the equation above implies $n\theta$ to be an even integral multiple of $2\pi$. Therefore, we conclude that the solutions of the equation (228), namely the $n$–**th roots of unity**, are:

$$1, e^{\frac{2\pi i}{n}}, e^{\frac{4\pi i}{n}}, \dots, e^{\frac{2(n-1)\pi i}{n}} \tag{229}$$

There's no need to go beyond the $(n-1)$–th multiple of $\frac{2\pi}{n}$, since then we will just recover the same numbers over again (this is simply because $\theta$ and $\theta + 2\pi$ represent the same angle).